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Time Speed & Distance

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TIME SPEED & DISTANCE

The topic starts with the basic formula i.e. **Speed** = $\frac{\text{distance}}{\text{time}}$

From this we can derive the following:

$$\text{Time} = \frac{\text{distance}}{\text{speed}} \quad \text{Distance} = \text{Speed} \times \text{Time}$$

A. Units

The two units that are normally taken **kilometer** per hour (km/h) and meter per second (m/s) $1 \text{ km/hr} = \left(\frac{1 \times 1000}{1 \times 3600} \right) = \frac{5}{18} \text{ m/sec.}$

For conversion of speed from **km/hr to m/sec**, **multiply** the speed by **5/18**.

For conversion of speed from **m/sec to km/h**, **multiply** the speed by **18/5**.

B. Solved Examples

Ex 1: What is the distance covered by a car (in km) traveling at a speed of 84 km/hr in 10 minutes?

Sol. Distance covered = $84 \times \frac{10}{60} = 14 \text{ km.}$

(Time is to taken in hours as speed is km/h).

Ex 2: A bus is moving at a speed of 100 km/hr, how long (in min) is it going to take to travel 25 km?

Sol. Distance = Speed $\times$ Time $\therefore$ Time = Distance $\div$ Speed



$$\Rightarrow \frac{25}{100} = 0.25 \text{ hours} = 15 \text{ min.}$$

Note: While converting decimal hours into the minutes, these are to multiplied with 60 and not by hundred.

Ex 3: A man covers 75 km in 90 minutes. What is his speed in km/h?

Sol. Speed = Distance $\div$ Time.

Since, time is given in minutes and required answer is in km/h, so we need to convert time into equivalent hours.

$$90 \text{ minutes} = \frac{90}{60} = 1.5 \text{ hours.}$$

$$\text{Therefore, speed} = \frac{75}{1.5} = 50 \text{ km/h.}$$

C. Ratio based TSD concept

The concept of ratio is a very important tool while solving questions in any aptitude-based paper. A lot of calculations can be avoided by applying basic concepts of ratio. In this topic, we will learn to apply ratio to solve Time, Speed and Distance (TSD) questions.

D. Basic Concept

The basic formula says Speed = Distance/Time. From this formula, it is evident that speed is directly proportional to distance and inversely proportional to time.

Basically, it means if two objects have their speeds in the ratio a:b, then distance covered by the two objects in same time



will be in the ratio $a:b$ and time taken by the two objects to cover same distance will be in the ratio $b:a$.

Ex 4: Walking $\frac{3}{5}$ th of his usual speed, Ravi reached his destination 20 minutes late. Find his usual time.

Sol. $\frac{3}{5}$ of usual speed implies $\frac{5}{3}$ of usual time (ut)

$$\frac{5}{3} \text{ of ut} = \text{ut} + 20 \text{ min} \Rightarrow \frac{5}{3} \text{ of ut} - \text{ut} = 20 \text{ min}$$

$$\Rightarrow \frac{2}{3} \text{ of ut} = 20 \text{ min} \Rightarrow \text{ut} = 30 \text{ min}$$

Although this question has been solved by making a simple equation, still the use of ratio can eliminate the calculation-work. So, let's have an example where the ratio concept has been used.

Ex 5: If I travelled at $\frac{4}{3}$ rd of my usual speed, I reached my destination 20 minutes early. What is the time that I usually take to reach my destination?

Sol: Since, the current speed is $\frac{4}{3}$ rd the usual speed, so we can take the current speed as 4 km/hr and the usual speed as 3 km/hr. Hence the ratio of the speeds is 4:3.

As the distance covered is same in both the cases, so the time taken will be in the ratio 3:4 (in the reverse ratio of speeds).

It means that if today, I took 3 minutes, then on usually I take 4 minutes i.e. I am one minute early.



Now apply simple chain rule that, if I am early by 1 minute, the usual time taken is 4 minutes and if I am 20 minutes early, the normal time taken will be $20 \times 4 = 80$ minutes.

Ex 6: A thief flees city A in a car towards city B on a stretch of straight road, 600 km long, at a speed of 50 km/hr. In 1 hour, a police party leaves city A to chase the thief at 60 km/hr. Find the distance travelled by the police when it catches thief.

Sol: In this question the thief travelled for the first 1 hour. The speed of thief is 50 km/hr, so he has covered 50 km in 1 hour.

We can see that the ratio of speeds of thief and police is 50: 60 or 5: 6.

Excluding the first 1 hour, the thief and the police have run for the same time. So, the distance covered is in the ratio of their speeds i.e. 5: 6.

Since, the initial distance between the thief and the police is 50 km, so in total, thief and police have covered 250 km and 300 km respectively (to make the ratio 5:6).

E. Average Speed

(i) When distance is same for two different speeds

Let the two speeds be a km/hr and b km/hr.

Let the distance traveled at each of these speeds be x km.

Time taken to cover x km at ' a ' km/hr = $\frac{x}{a}$ & at ' b ' km/hr = $\frac{x}{b}$.

$$\text{Total time taken} = \frac{x}{a} + \frac{x}{b} = \frac{bx + ax}{ab} = \frac{x(b + a)}{ab},$$

And the total distance covered = $2x$.

$$\text{Therefore, average speed} = \frac{2x}{\frac{x(a + b)}{ab}} = \frac{2ab}{a + b}.$$

(ii) When time is same for two different speeds

The average speed of traveling at two different speeds for the same time duration is just the simple average of these two speeds. Average speed = $(a + b)/2$

Ex 7: While travelling from home to school, a student travelled at the speed of 75 km/hr. While coming back home from school on the same path, he travelled at the speed of 25 km/hr. What is my average speed of travel?

Sol. Average speed = $\frac{2ab}{a + b} = (2 \times 75 \times 25)/100 = 37.5 \text{ km/hr}$

Ex 8: A motorist travels for one hour at an average speed of 60 km/hr and next hour at an average speed of 70 km/hr. Then his average speed is

Sol. Average speed = $(60 + 70) \div 2 = 65 \text{ km/hr}$.

Ex 9: An aeroplane travels distances of 2500 km, 1200 km and 500 km at speeds of 500 km/hr, 400 km/hr and 250 km/hr respectively. The average speed is

A) 420 km/hr

B) 405 km/hr

C) 410 km/hr

D) 575 km/hr



Sol. Total distance = $2500 + 1200 + 500 = 4200$ km. Total

$$\text{time taken} = \frac{2500}{500} + \frac{1200}{400} + \frac{500}{250} = 10 \text{ hrs}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}} = \frac{4200}{10} = 420 \text{ km/hr}$$

F. Relative Speed

Let the speed of two objects be x and y

(i) In same direction, their relative speed is the difference between the two speeds

$$\text{Relative speed} = x - y \text{ (considering } x > y)$$

(ii) In opposite direction, their relative speed is the sum of the two speeds.

$$\text{Relative speed} = x + y$$

Some more facts:

- (i) If a train crosses a stationary man / lamp post / sign post /** then the distance travelled is the **length of the train**.
- (ii) If a train crosses a platform / bridge** then the distance traveled is the **length of the train + length of the object**.
- (iii) If a train crosses a car / bicycle / a moving man** - in these cases, the relative speed between the train and the object is taken depending on the direction of movement of other object relative to the train and the distance traveled is the **length of the train**.



- (iv) If a train crosses another train which is moving at a particular speed in the same or opposite direction – in these cases, the other train is moving as well and relative speed between them is taken depending on the direction of the other train and distance travelled is the **sum of the lengths of both the trains**.

G. Some More Solved Examples

Ex 10: A train traveling at 60 km/hr crosses a man in 6 sec. Calculate the length of the train in meters.

Sol. Speed in m/sec = $60 \times \frac{5}{18} = \frac{50}{3}$ m/sec.

Time taken to cross the man = 6 seconds.

Hence distance travelled = $\frac{50}{3} \times 6 = 100$ meters = length of the train.

Ex 11: A train traveling at 60 km/hr crosses another train traveling in the same direction at 50 km/hr in 45 sec. What is the total combined length of both the trains?

Sol. Speed of train A in m/sec = $60 \times \frac{5}{18} = \frac{50}{3}$ m/sec.

Speed of train B in m/sec = $50 \times \frac{5}{18} = \frac{125}{9}$ m/sec

The relative speed is = $\frac{50}{3} - \frac{125}{9} = \frac{25}{9}$ m/sec.

Time taken for train A to cross train B = 45 seconds.



Therefore, distance travelled

$$= \frac{25}{9} \times 45 = 125 \text{ m} = \text{Combined length of two trains.}$$

Ex 12: A train crosses a platform of length 300 m in 30 seconds and overtakes a car traveling in the same direction at speed of 90 km/hr in 120 seconds. What is the length of train (in m) and the speed at which it is traveling (in km/hr)?

Sol. Let the length of the train = x m and speed of the train = s km/hr

We know, distance = speed $\times$ time

$$\text{Given, } (x + 300) = s \times \frac{5}{18} \times 30$$

$$\Rightarrow x = s \times \frac{5}{18} \times 30 - 300$$

Also, since the car is travelling in the same direction, $x = (s - 90) \times \frac{5}{18} \times 120$

$$\text{Solving, } s \times \frac{5}{18} \times 30 - 300 = s \times \frac{5}{18} \times 120 - 3000$$

$$\Rightarrow 25s = 2700 \Rightarrow s = 108 \text{ km/hr and } x = 600 \text{ m}$$

So, length of the train = 600 m and speed of the train = 108 km/hr

Ex 13: A train traveling at 108 km/hr crosses a bridge of 600 mt length completely in 30 seconds. What is the length of the train?

Sol. Speed = 108 km/hr = $108 \times \frac{5}{18} = 30$ m/sec. Time taken to cross = 30 seconds.



Therefore, distance traveled = $30 \times 30 = 900$ m.

Distance = Length of the train + length of the bridge

$900 = \text{Length of the train} + 600$, Length of the train = 300 m

Ex 14: A man is walking at the speed of 20 kmph. After every kilometer, he takes rest for 10 min. How much time will he take to cover a distance of 10 km?

A) 110 min B) 100 min C) 120 min D) 90 min

Sol. If a man covers 10 km with a speed of 20 kmph, he will take a total time of $10/20$ hrs or 30 min. But he rests after every km for 10 min, so in total 10 km distance, he will take 9 breaks.

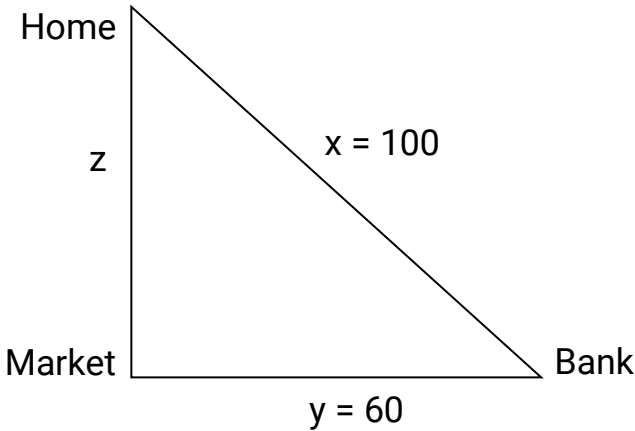
So, the total time spent in taking the rest is $9 \times 10 = 90$ min.

Hence, the total time spent to cover the distance of 10 km = 30 min + 90 min = 120 min or 2 hrs

Ex 15: Mr. Akshay stays in a triangular colony. He stays at one vertex, with a bank and a market at the other two vertices. The roads meeting at the market are at 90° . Akshay goes to the bank from his home and then to the market totally covering 160 m. He then walks straight back home from the market. If distance from home to bank is 40 m more than what he covered from bank to market, find the distance he walked from market to home.



Sol.



$$x + y = 160 \text{ and } x - y = 40$$

$$x = 100 \text{ \& } y = 60$$

$$x^2 = y^2 + z^2$$

$$\text{Thus } z = \sqrt{(100^2 - 60^2)} = \sqrt{6400} = 80 \text{ m}$$

Ex 16: 2 trains each of length 120 metres move in the same direction. The faster train completely overtakes the slower one in 15 seconds. If the slower train were to move at half its speed, faster train would overtake it in 10 seconds. Find the speeds of the two trains (faster and slower respectively in m/s)

Sol. The total distance to be covered by one train to completely overtake the other = Sum of the lengths of the 2 trains = $120 + 120 = 240$ metres.

Let the speed of the faster train be x m/s and that of the slower train y m/s.



In the first case, Relative Speed = $x - y$

(Since both are moving in the same direction)

$$\text{Relative speed} = \frac{\text{Distance}}{\text{Time taken}}$$

$$\therefore x - y = 240/15 = 16 \text{ m/s}$$

In the second case, the slower train moves at half its speed.

$$\text{Relative Speed} = x - 0.5y$$

(since both are moving in the same direction)

$$\text{Relative speed} = \frac{\text{Distance}}{\text{Time taken}}$$

$$\therefore x - 0.5y = 240/10 = 24 \text{ m/s.}$$

$$\text{Solving } x - y = 16 \text{ and } x - 0.5y = 24.$$

$$\text{Hence } x = 32 \text{ and } y = 16$$

$\therefore$ the speeds of the faster and slower trains are 32 m/s and 16 m/s respectively.

Ex 17: A woman is walking down a downward escalator and steps down 10 steps to reach the bottom. Just as she reaches the bottom of the escalator, a sale commences on the floor above. She runs back up the downward moving escalator at a speed 5 times that which she walked down. She covers 25 steps in reaching the top. How many steps are visible on the escalator when it is switched off?

A) 20

B) 15

C) 25

D) 10



Sol: Let the speed of the escalator is 's' steps per minute

Let the speed of woman be 1 step per minute.

$$\text{Hence, } 10 + 10s = 25 - 5s \Rightarrow s = 1$$

So, total number of steps in the escalator is $10 + 10 \times 1$
= 20 steps

Ex 18: A man travelled 2000 km in 5 hrs, partly by air and partly by train. Had he travelled all the way by air, he would have saved $3/4^{\text{th}}$ of the time he was in the train and would have arrived 3 hrs early. Find the distance he travelled by air and train.

Sol: Total distance travelled = 2000 km. Total time taken = 5 hrs.

$$\text{Now } 3/4 \times (\text{Total time in train}) = 3$$

$$\Rightarrow \text{Total time in train} = 4 \text{ hrs.}$$

$$\text{Total time in air} = 5 - 4 = 1 \text{ hr.}$$

So in 2 hrs, he would have covered 2000 km by air.

$$\text{Hence speed by air} = 2000/2 = 1000 \text{ kmph.}$$

$$\text{So total distance in air} = 1000 \times 1 = 1000 \text{ km.}$$

Hence distance travelled by train

$$= 2000 - 1000 = 1000 \text{ km.}$$